

On the γ -Spreadness of Average-Case to Worst-Case Transformations

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August 21, 2025

Abstract

Achieving (at least) a worst-case correctness error is essential for an underlying public-key encryption (PKE) scheme to which the Fujisaki-Okamoto (FO) transformation is applied. There are three average-case to worst-case (ACWC) transformations—denoted as ACWC_0 , ACWC_1 (PKC 2023), and ACWC_2 (TIFS 2023)—which generically convert a PKE scheme with an average-case correctness error into one with a worst-case correctness error. However, in these ACWC transformations the γ -spreadness, a critical factor in determining explicit rejection (FO^+) or implicit rejection ($\text{FO}^\perp$), has not been established with rigorous proofs. Existing analyses of γ -spreadness lack rigorous proofs, include analytical flaws, or fail to achieve the tightest possible bounds. In this work, we reprove the γ -spreadness of ACWC-transformed PKE schemes by leveraging two key facts: the random oracle is chosen at random and the encoding mechanism used in the ACWC framework is message-hiding. Our new proofs are applied to the previous NTRU-based PKE schemes, called NTRU-C, NTRU-B, and NTRU+, giving the corrected γ -spreadness for those PKE schemes with concrete parameters.

Keywords: γ -spreadness, ACWC transformation, Fujisaki-Okamoto transformation, NTRU

1 Introduction

A family of variants [3, 9] of the Fujisaki-Okamoto transformation [6, 7] has been developed to instantiate a (post-quantum) key encapsulation mechanism (KEM) that is secure against chosen-ciphertext attacks (i.e., IND-CCA-secure) from any public-key encryption (PKE) scheme that satisfies chosen-plaintext security or one-wayness (i.e., $\{\text{IND}, \text{OW}\}$ -CPA-secure). A sequence of works [9, 2, 15, 4, 10, 8, 12] has improved the security reduction for the PKE-to-KEM transformation (denoted FO) in the quantum random oracle model (QROM). Beyond the base $\{\text{IND}, \text{OW}\}$ -CPA security, any PKE scheme (denoted PKE') used as the underlying scheme for FO must achieve (at least) a negligible *worst-case* correctness error. Roughly speaking, a PKE scheme is said to be worst-case δ' -correct if, for any message and any honestly sampled randomness (for a prescribed distribution), the maximum probability of decryption failure is bounded by δ' .

To systematically provide a negligible worst-case correctness error for an underlying PKE' , three average-case to worst-case (ACWC) transformations—denoted ACWC_0 [5], ACWC_1 [5], and ACWC_2 [13] have been proposed. Let PKE be an underlying PKE scheme whose decryption error δ is negligible in the *average*

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case, meaning that for any honestly sampled message and randomness (for prescribed distributions) the maximum probability of decryption failure is bounded by some negligible δ . Each ACWC transformation produces $\text{PKE}' = \text{ACWC}_i[\text{PKE}]$ for $i = 0, 1, 2$, while preserving or improving the security of PKE. A benefit of these transformations is that the resulting worst-case correctness error δ' of PKE' remains essentially the same as the average-case correctness error δ of PKE, so that one only needs to analyze failures for honestly sampled inputs to control the worst-case correctness error of PKE' .

In the decapsulation algorithm of an FO-transformed KEM, there are two types of rejections in response to an invalid ciphertext: explicit rejection ($\text{FO}^\perp$), which outputs the error symbol $\perp$, or implicit rejection ($\text{FO}^\times$), which proceeds to output a pseudorandom key. The choice between them depends on whether the underlying PKE scheme PKE' satisfies the γ -spreadness. Roughly speaking, the γ -spreadness means that for any fixed message m and ciphertext c , the maximum over randomnesses of the probability that $c = \text{Enc}(pk, m; r)$ is less than $2^{-\gamma}$. If PKE' satisfies the γ -spreadness, the explicit rejection is permissible; otherwise, one must fall back to the implicit rejection. From an efficiency perspective, the explicit rejection is preferable, because the implicit rejection always needs to compute the pseudorandom key, regardless of whether a ciphertext is valid or not [1, 16]. In terms of security, [2, 11] show that the explicit rejection implies the implicit rejection, but the reverse direction incurs security loss and only holds if PKE' satisfies the γ -spreadness.

The three ACWC transformations present generic ways to provide for the γ -spreadness of PKE' . [5, 13] claim that if the underlying scheme PKE is γ -spread, then so is PKE' , *without rigorous proofs*. Although the ACWC constructions successfully control worst-case correctness error, their internal structure complicates a direct application of the standard γ -spreadness analysis. Typically, one fixes a message m and ciphertext c (under a key pair (pk, sk)) and considers the maximum probability that is summed over all possible r such that $c = \text{Enc}(pk, m; r)$. This typical analysis cannot be directly applied to the γ -spreadness of PKE' . For instance, during ACWC_2 , a fixed message m (relative to PKE') is encoded as $M := \text{Encode}(m, G(r))$ using some encoding method ‘Encode’ and a hash function G . Then, M is encrypted as $c = \text{Enc}(pk, M; r)$. Consequently, even for a fixed m , as r varies according to its distribution, the encoded plaintext M also changes, so M cannot be treated as fixed in the usual analysis. Analogous complications arise in ACWC_0 and ACWC_1 , which require new rigorous proofs on the γ -spreadness of PKE' when applying those ACWC transformations.

1.1 Our result

Assuming that PKE is γ -spread, we revisit the γ' -spreadness of PKE' obtained by applying the ACWC transformations to PKE. The previous proofs on the γ' -spreadness of PKE' under ACWC_0 and ACWC_1 were presented without sufficiently reflecting our argument mentioned above. In particular, the proof under ACWC_0 was incorrect, simply asserting that $\gamma' = \gamma$. Also, by considering our argument, [14] presented a proof on the γ' -spreadness of PKE' under ACWC_2 , but the resulting bound on the γ' -spreadness can be further improved.

We reprove the γ' -spreadness of PKE' under ACWC_0 , ACWC_1 , and ACWC_2 , respectively. All our proofs are based on some encoding method Encode and a hash function G modeled as a random oracle. Encode takes as input a message m (relative to PKE') and a hash value, and has a so-called *message-hiding* property that the output of Encode is distributed independently of m . Again, let (pk, sk) be a pair of public and private keys for PKE. Using Encode and G , all ciphertexts of PKE' share the following similarity: $c = (c_1, c_2) = (\text{Enc}(pk, M; r), \text{Encode}(m, G(M)))$ under ACWC_0 , $c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r)$ under ACWC_1 , and $c = \text{Enc}(pk, \text{Encode}(m, G(r)); r)$ under ACWC_2 . Given randomness R relative to PKE' , R is used to sample (M, r) under ACWC_0 , (M_1, r) under ACWC_1 , and r under ACWC_2 , following some

distributions, respectively. Thus, we can see that each encrypted message (relative to PKE) also changes as a new randomness R is chosen. As mentioned above, this is the difficulty in directly analyzing the γ' -spreadness of PKE' . Then, our proof strategy is as follows:

1. Given a ciphertext $c = (c_1, c_2) = (\text{Enc}(pk, M; r), \text{Encode}(m, G(M)))$ under ACWC_0 , the second component c_2 is distributed independently of m because of the message-hiding property of Encode . Next, using the fact that G is chosen at random in the random oracle model, c_2 can be viewed as random choice from the output space of Encode , indicating that the distributions of c_1 and c_2 are independent. Then, we can consider (M, r, c_2) as being sampled according to some distributions, in which case we adapt the Law of Total Probability to deal with all possible choices of (M, r, c_2) .
2. Given a ciphertext $c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r)$ under ACWC_1 , the encoded partial message $M_2 := \text{Encode}(m, G(M_1))$ is distributed independently of m because of the message-hiding property of Encode . Additionally, the random choice of G allows for considering M_2 as being sampled from some distribution. Similarly to ACWC_0 , we can then use the Law of Total Probability to address all possible choices of $(M_1 || M_2, r)$.
3. Given a ciphertext $c = \text{Enc}(pk, \text{Encode}(m, G(r)); r)$ under ACWC_2 , the encoded message $M := \text{Encode}(m, G(r))$ is distributed independently of m because of the message-hiding property of Encode . Additionally, the random choice of G allows for considering M as being sampled according to some distribution. Similarly to ACWC_0 and ACWC_1 , we can then use the Law of Total Probability to address all possible choices of (M, r) .

All possible cases are rearranged into a group of encryptions, in which a fixed message M and all possible randomnesses $\{r\}$ are involved. For each group, we now apply the γ -spreadness of PKE to upper-bound the γ' -spreadness of PKE' , when a specific ciphertext c of PKE' is given. In case of ACWC_0 , we additionally consider the probability that a sampled c_2 becomes the second component of the given c . Then, our new proof shows that $\gamma' = \gamma + \lambda$, where the output space of Encode under ACWC_0 is instantiated with a set of λ -bit strings. In case of ACWC_1 and ACWC_2 , we prove that $\gamma' = \gamma$. Although we follow a similar proof strategy [14], our result under ACWC_2 gives a tighter and more precise γ' -spreadness of PKE' .

Finally, we recompute the γ' -spreadness of three previous NTRU-based PKE's, which are NTRU-C [5], NTRU-B [5], and NTRU+ [14]. All are based on NTRU-based PKE schemes (as PKE) that are constructed over rings with cyclotomic trinomials. As noted in [14], it is easy to analyze the γ -spreadness of an NTRU-based PKE scheme in a statistical manner, given some distributions for sampling coefficients of a polynomial. By applying the three ACWC transformations to an NTRU-based PKE, we can have NTRU-C via ACWC_0 , NTRU-B via ACWC_1 , and NTRU+ via ACWC_2 , respectively. During such transformations, our new proofs give the corrected γ' -spreadness of the resultant NTRU-based PKE's, as shown in Table 1.

2 Preliminaries

2.1 Notation

The set $\mathbb{Z}_q$ is defined as $\{-(q-1)/2, \dots, (q-1)/2\}$, where q is a positive odd integer. For an integer $a \in \mathbb{Z}$, a modulo q is congruent to a unique integer $x \in \mathbb{Z}_q$ such that $q \mid (x - a)$. The polynomial ring R_q is defined as $\mathbb{Z}_q[x] / \langle f(x) \rangle$. If X is a finite set and D is a distribution, sampling is denoted as follows: the notation $a \leftarrow X$ indicates that a is selected uniformly at random from the set X , while $a \leftarrow D$ denotes that a is sampled randomly from the distribution D . Additionally, $\mathbf{a} \leftarrow D$ means that the coefficients of the

polynomial a are sampled independently according to the distribution D . We denote $\llbracket B \rrbracket$ as 1 if the Boolean statement B is true, and 0 otherwise.

2.2 Definitions

Definition 2.1 (Public-Key Encryption). A public key encryption scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$, randomness space $\mathcal{R}$, and ciphertext space $\mathcal{C}$ consists of the following three algorithms:

- $\text{Gen}(1^n)$: The key generation algorithm Gen is a probabilistic algorithm that takes a security parameter 1^n as input and outputs a pair of public and secret keys (pk, sk) .
- $\text{Enc}(pk, m; r)$: The encryption algorithm Enc is a probabilistic algorithm that takes a public key pk , a message $m \in \mathcal{M}$, and randomness $r \in \mathcal{R}$ as input and outputs a ciphertext $c \in \mathcal{C}$. If we write $\text{Enc}(pk, m)$, it is to denote the encryption algorithm without explicitly mentioning the randomness.
- $\text{Dec}(sk, c)$: The decryption algorithm Dec is a deterministic algorithm. Dec takes a secret key sk and a ciphertext $c \in \mathcal{C}$ as input and outputs either a message $m \in \mathcal{M}$ or a special symbol $\perp \notin \mathcal{M}$ to indicate that c is an invalid ciphertext.

Spreadness. PKE is γ -spread [4] if

$$\min_{m \in \mathcal{M}, (sk, pk)} \left(-\log \max_{c \in \mathcal{C}} \Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, m; r)] \right) \geq \gamma,$$

where the probability is over the randomness $r \leftarrow \psi_{\mathcal{R}}$ and the minimum is over all key pairs (pk, sk) generated by Gen . This definition can be relaxed by considering an expectation over the choice of (pk, sk) . PKE is *weakly* γ -spread [4] if

$$-\log \mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, m; r)] \right] \geq \gamma,$$

where the probability is over the randomness $r \leftarrow \psi_{\mathcal{R}}$ and the expectation is over $(pk, sk) \leftarrow \text{Gen}(1^n)$.

Correctness. PKE is worst-case correctness error δ [9] if

$$\mathbb{E} \left[\max_{m \in \mathcal{M}} \Pr_{r \leftarrow \psi_{\mathcal{R}}} [\text{Dec}(sk, \text{Enc}(pk, m; r)) \neq m] \right] \leq \delta,$$

where the expectation is over $(pk, sk) \leftarrow \text{Gen}(1^n)$ and the choice of the random oracles (if any).

PKE is average-case correctness error δ if

$$\mathbb{E} \left[\Pr_{m \leftarrow \psi_{\mathcal{M}}, r \leftarrow \psi_{\mathcal{R}}} [\text{Dec}(sk, \text{Enc}(pk, m; r)) \neq m] \right] \leq \delta,$$

where the expectation is over $(pk, sk) \leftarrow \text{Gen}(1^n)$ and the choice of the random oracles (if any).

Randomness Recoverability. PKE is defined as *randomness recoverable* (RR) if there exists an algorithm RRec such that for all key pairs $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$, and for any message $m \in \mathcal{M}$ and ciphertext $c \in \mathcal{C}$, the following condition holds:

$$\text{if } r = \text{RRec}(pk, m, c) \in \mathcal{R}, \text{ then } c = \text{Enc}(pk, m; r).$$

3 ACWC Transformations

We describe three average-case to worst-case transformations, namely ACWC_0 [5], ACWC_1 [5] and ACWC_2 [14]. Each transformation takes an underlying PKE with average-case correctness error δ and produces a derived scheme PKE' with worst-case correctness error δ' , satisfying $\delta' \approx \delta$.

3.1 ACWC_0

Figure 1 presents ACWC_0 and the instantiation $\text{PKE}' = \text{ACWC}_0[\text{PKE}, \text{G}]$ where $\text{G} : \{0, 1\}^* \rightarrow \mathcal{U}$ for the distribution $\psi_{\mathcal{U}}$. In particular, ACWC_0 assumes $\mathcal{U} = \{0, 1\}^\lambda$ for a security parameter λ , and $\psi_{\mathcal{U}}$ is the uniform distribution over $\{0, 1\}^\lambda$. The derived scheme $\text{PKE}' = (\text{Gen}', \text{Enc}', \text{Dec}')$ has message space $\mathcal{M}' = \{0, 1\}^\lambda$ and randomness space $\mathcal{R}'$, while the base scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ has message space $\mathcal{M}$ and randomness space $\mathcal{R}$ with respective distributions $\psi_{\mathcal{M}}$ and $\psi_{\mathcal{R}}$. In terms of security, [5] proves that PKE' is secure against chosen-plaintext attacks (i.e., IND-CPA secure) if PKE is one-way against chosen-plaintext attacks (i.e., OW-CPA secure), assuming G is modeled as a (quantum) random oracle. ACWC_0 is conceptually simple and transparent, but its instantiation increases ciphertext length by λ bits.

$\text{Gen}'(1^n)$	$\text{Dec}'(sk, c)$
1: $(pk, sk) := \text{Gen}(1^n)$	1: $M := \text{Dec}(sk, c_1)$
2: return (pk, sk)	2: $m := \text{G}(M) \oplus c_2$
	3: return m
$\text{Enc}'(pk, m \in \mathcal{M}'; (R_1, R_2) \in \mathcal{R}')$	
1: $M \leftarrow \psi_{\mathcal{M}}$ using the randomness R_1	
2: $r \leftarrow \psi_{\mathcal{R}}$ using the randomness R_2	
3: $c_1 := \text{Enc}(pk, M; r)$	
4: $c_2 := \text{G}(M) \oplus m$	
5: $c := (c_1, c_2)$	
6: return c	

Figure 1: $\text{ACWC}_0[\text{PKE}, \text{G}]$

3.2 ACWC_1

The transformation ACWC_1 employs, in addition to a hash function G modeled as a random oracle, a supplementary primitive called a generalized one-time pad (GOTP) [5].

Definition 3.1. A generalized one-time pad $\text{GOTP} = (\text{Encode}, \text{Inv})$ with a message space $\mathcal{X}$, a randomness space $\mathcal{U}$, and a code space $\mathcal{Y}$ (with distributions $\psi_{\mathcal{X}}, \psi_{\mathcal{U}}, \psi_{\mathcal{Y}}$) consists of the following deterministic algorithms:

- $\text{Encode}(x, u)$: on input a message $x \in \mathcal{X}$ and random $u \in \mathcal{U}$ outputs a code $y \in \mathcal{Y}$.
- $\text{Inv}(y, u)$: on input a code $y \in \mathcal{Y}$ and random $u \in \mathcal{U}$ recovers a message $x \in \mathcal{X}$.

It satisfies the following properties:

1. **Decoding:** $\text{Inv}(\text{Encode}(x, u), u) = x$ for all $x \in \mathcal{X}, u \in \mathcal{U}$.

2. **Message hiding:** For every $x \in \mathcal{X}$, the distribution of $\text{Encode}(x, u)$ for $u \leftarrow \psi_{\mathcal{U}}$ equals $\psi_{\mathcal{Y}}$.
3. **Randomness hiding:** For every $u \in \mathcal{U}$, the distribution of $\text{Encode}(x, u)$ for $x \leftarrow \psi_{\mathcal{X}}$ equals $\psi_{\mathcal{Y}}$.

For ACWC_1 , the derived scheme $\text{PKE}' = (\text{Gen}', \text{Enc}', \text{Dec}')$ has message space $\mathcal{M}'$ and randomness space $\mathcal{R}'$, while the base scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ has message space $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2$ with distribution $\psi_{\mathcal{M}} = \psi_{\mathcal{M}_1} \times \psi_{\mathcal{M}_2}$, and randomness space $\mathcal{R}$ with distribution $\psi_{\mathcal{R}}$. Let $G : \{0, 1\}^* \rightarrow \mathcal{U}$ for the distribution $\psi_{\mathcal{U}}$, and let $\text{Encode} : \mathcal{M}' \times \mathcal{U} \rightarrow \mathcal{M}_2$ and $\text{Inv} : \mathcal{M}_2 \times \mathcal{U} \rightarrow \mathcal{M}'$ be the GOTP algorithms. Figure 2 illustrates ACWC_1 and the instantiation $\text{PKE}' = \text{ACWC}_1[\text{PKE}, \text{GOTP}, G]$. By the message-hiding property of GOTP, the encoded component M_2 is distributed as $\psi_{\mathcal{M}_2}$ independently of the choice of $m \in \mathcal{M}'$. Regarding security, [5] establishes that, in the (Q)ROM with G as a random oracle, PKE' is one-way under chosen-plaintext attacks (i.e., OW-CPA secure) whenever PKE is so. Unlike ACWC_0 , ACWC_1 does not require adding redundancy to ciphertexts.

<u>$\text{Gen}'(1^n)$</u>	<u>$\text{Dec}'(sk, c)$</u>
1: $(pk, sk) := \text{Gen}(1^n)$	1: $M_1 M_2 := \text{Dec}(sk, c)$
2: return (pk, sk)	2: $m := \text{Inv}(M_2, G(M_1))$
	3: return m
<u>$\text{Enc}'(pk, m \in \mathcal{M}'; (R_1, R_2) \in \mathcal{R}')$</u>	
1: $M_1 \leftarrow \psi_{\mathcal{M}_1}$ using the randomness R_1	
2: $r \leftarrow \psi_{\mathcal{R}}$ using the randomness R_2	
3: $M_2 := \text{Encode}(m, G(M_1))$	
4: $c := \text{Enc}(pk, M_1 M_2; r)$	
5: return c	

Figure 2: $\text{ACWC}_1[\text{PKE}, \text{GOTP}, G]$

3.3 ACWC_2

Analogously to ACWC_1 , ACWC_2 employs a semi-generalized one-time pad (SOTP) [13] together with a hash function G modeled as a (quantum) random oracle.

Definition 3.2. A semi-generalized one-time pad $\text{SOTP} = (\text{Encode}, \text{Inv})$ with a message space $\mathcal{X}$, a randomness space $\mathcal{U}$, and a code space $\mathcal{Y}$ (with distributions $\psi_{\mathcal{X}}, \psi_{\mathcal{U}}, \psi_{\mathcal{Y}}$) comprises deterministic algorithms:

- $\text{Encode}(x, u)$: on input a message $x \in \mathcal{X}$ and a randomness $u \in \mathcal{U}$ outputs a code $y \in \mathcal{Y}$.
- $\text{Inv}(y, u)$: on input a code $y \in \mathcal{Y}$ and a randomness $u \in \mathcal{U}$ outputs either a message $x \in \mathcal{X}$ or $\perp$.

It satisfies:

1. **Decoding:** $\text{Inv}(\text{Encode}(x, u), u) = x$ for all $x \in \mathcal{X}, u \in \mathcal{U}$.
2. **Message hiding:** For every $x \in \mathcal{X}$, the distribution of $\text{Encode}(x, u)$ with $u \leftarrow \psi_{\mathcal{U}}$ equals $\psi_{\mathcal{Y}}$.
3. **Rigidity:** For all $u \in \mathcal{U}$ and $y \in \mathcal{Y}$ with $\text{Inv}(y, u) \neq \perp$, one has $\text{Encode}(\text{Inv}(y, u), u) = y$.

There are two differences between GOTP and SOTP: (1) Inv of SOTP outputs $\perp$ for an invalid input, which means that there is a process to determine whether the input (y, u) is valid or not, and (2) SOTP needs the rigidity, instead of the randomness-hiding property of GOTP.

For ACWC₂, the derived PKE' = (Gen', Enc', Dec') has message space $\mathcal{M}'$ and randomness space $\mathcal{R}'$, while the underlying PKE = (Gen, Enc, Dec, RRec) has message space $\mathcal{M}$ and randomness space $\mathcal{R}$ with distributions $\psi_{\mathcal{M}}$ and $\psi_{\mathcal{R}}$. The additional randomness-recovery algorithm RRec guarantees that given a ciphertext c and its decryption $M \in \mathcal{M}$, one can recover the associated randomness $r \in \mathcal{R}$ such that $c = \text{Enc}(pk, M; r)$; if $r \notin \mathcal{R}$ exists, RRec outputs $\perp$, implicitly checking validity. Let $G : \{0, 1\}^* \rightarrow \mathcal{U}$ for the distribution $\psi_{\mathcal{U}}$, and let $\text{Encode} : \mathcal{M}' \times \mathcal{U} \rightarrow \mathcal{M}$ and $\text{Inv} : \mathcal{M} \times \mathcal{U} \rightarrow \mathcal{M}'$ be the SOTP algorithms. Figure 3 illustrates ACWC₂ and the instantiation PKE' = ACWC₂[PKE, SOTP, G]. By the message-hiding property of SOTP, the encoded M is distributed as $\psi_{\mathcal{M}}$ regardless of the choice of $m \in \mathcal{M}'$. Regarding security, [13] establishes that in the (Q)ROM, PKE' is IND-CPA secure assuming PKE is OW-CPA secure¹.

<u>Gen'(1^λ)</u>	<u>Dec'(sk, c)</u>
1: $(pk, sk) := \text{Gen}(1^\lambda)$	1: $M := \text{Dec}(sk, c)$
2: return (pk, sk)	2: $r := \text{RRec}(pk, M, c)$
	3: $m := \text{Inv}(M, G(r))$
<u>Enc'(pk, m ∈ M'; R ∈ R')</u>	4: if $r \notin \mathcal{R}$ or $m = \perp$
1: $r \leftarrow \psi_{\mathcal{R}}$ using the randomness R	5: return $\perp$
2: $M := \text{Encode}(m, G(r))$	6: return m
3: $c := \text{Enc}(pk, M; r)$	
4: return c	

Figure 3: ACWC₂[PKE, SOTP, G]

4 Spreadness Proofs in ACWC Transformations

We show that an ACWC-transformed PKE' is γ' -spread, assuming that an underlying PKE is γ -spread and G is modeled as a random oracle. Our new rigorous analysis establishes $\gamma' = \gamma + \lambda$ for ACWC₀ and $\gamma' = \gamma$ for ACWC₁ and ACWC₂, meaning that all three ACWC transformations preserve the γ -spreadness of PKE well in the γ' -spreadness of PKE'. By contrast, previous analyses [5, 13] roughly stated that γ -spreadness lifts to PKE', neglecting the subtle dependency described below. Let $\{\mathcal{M}', \mathcal{C}', \mathcal{R}'\}$ be the {message, ciphertext, randomness} sets of PKE', respectively. In analyzing the γ' -spreadness of PKE', we fix $m \in \mathcal{M}'$ and consider the probability $\Pr[c = \text{Enc}'(pk, m; R)]$ over $c \in \mathcal{C}'$ and $R \leftarrow \psi_{\mathcal{R}'}$; the difficulty lies in the fact that, over random choices $R \leftarrow \psi_{\mathcal{R}'}$, a fixed message $m \in \mathcal{M}'$ of PKE' does not lead to a fixed message $M \in \mathcal{M}$ of PKE, and rather through all the above-mentioned ACWC transformations, M of PKE is changed according to a chosen randomness $R \in \mathcal{R}'$ of PKE. Therefore, we need to consider all possible randomness-dependent messages $\{M\}$ over random choices $R \leftarrow \psi_{\mathcal{R}'}$.

¹Their analysis in [13] additionally requires the base PKE to be message-recoverable and (computationally) injective; we omit those auxiliary assumptions here to focus on ACWC₂ and γ -spreadness.

4.1 New Proof on the γ' -Spreadness of $\text{PKE}' = \text{ACWC}_0[\text{PKE}, \text{G}]$

Let $m \in \mathcal{M}' = \{0, 1\}^\lambda$ be fixed. The γ' -spreadness of PKE' is defined as the maximum probability of sampling all randomnesses $R \leftarrow \psi_{\mathcal{R}}$ that $\text{Enc}'(pk, m; R)$ equals a given ciphertext $c \in \mathcal{C}'$. Under ACWC_0 , $c = (c_1, c_2)$ with $c_1 = \text{Enc}(pk, M; r)$ and $c_2 = G(M) \oplus m$, where (M, r) are sampled from $R = (R_1, R_2)$, respectively. Hence although m of PKE' is fixed, the underlying message $M \in \mathcal{M}$ of PKE varies with R . To reduce γ' to the γ -spreadness of PKE , we need to fix a message $M \in \mathcal{M}$ of PKE . To relate γ' to the γ -spreadness of PKE , we use the Law of Total Probability to consider all possible cases generated by all sampling messages $\{M_j\}$ and randomnesses $\{r_i\}$. Next, we leverage the random oracle property to ensure that $c_1 = \text{Enc}(pk, M_j; r_i)$ and $c_2 = G(M_j) \oplus m$ are probabilistically independent. Assuming that G is modeled as a random oracle and *chosen at random*, evaluating $G(M_j)$ is identical to sampling a random u from $\{0, 1\}^\lambda$, and by the message-hiding property of the One-Time Pad $c_2 = u \oplus m$ is also uniformly random in $\{0, 1\}^\lambda$. Therefore, by replacing $c_2 = G(M_j) \oplus m$ with a uniformly sampled $\tilde{c} \leftarrow \{0, 1\}^\lambda$, we can eliminate any dependence of c_2 on M_j , allowing us to treat c_1 and c_2 as independent in terms of their probability distributions. Lastly, we define two indicator functions $\mathbf{I}(pk, M_j, r_i, c_1)$ and $\mathbf{I}'(\tilde{c}, c_2)$ that output 0 or 1, reflecting the probability that $c_1 \stackrel{?}{=} \text{Enc}(pk, M_j; r_i)$ for a given (c_1, M_j, r_i) and the probability that $c_2 \stackrel{?}{=} \tilde{c}$ for a given $(c_2, \tilde{c})$. Using these functions, we can scan all possible combinations of (M_j, r_i) and $\tilde{c}$ that match a specific $c = (c_1, c_2)$. Especially, the sum of the indicator function $\mathbf{I}(pk, M_j, r_i, c_1)$ for all sampled $\{r_i\}$ is related to the γ -spreadness of PKE in case when M_j of PKE is fixed, because the function accounts only for the probability of sampling r_i such that $c_1 = \text{Enc}(pk, M_j; r_i)$. Using this property, we can upper-bound the sum of $\mathbf{I}(pk, M_j, r_i, c_1)$ for all sampled $\{r_i\}$ by the probability $2^{-\gamma}$ for each message M_j and a specific c_1 .

Theorem 4.1. If PKE is γ -spread and G is modeled as a random oracle, then $\text{PKE}' = \text{ACWC}_0[\text{PKE}, G]$ is γ' -spread with

$$\gamma' = \gamma + \lambda,$$

where λ is the message length of PKE' .

Proof. c is composed of (c_1, c_2) , and thus we need to consider the spreadness of both c_1 and c_2 in analyzing the spreadness of c . For a fixed key pair (pk, sk) and a fixed m in PKE' , we consider the probability $\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)]$ for every possible ciphertext c where $R = (R_1, R_2)$. Since M and r are sampled according to $\psi_{\mathcal{M}}$ and $\psi_{\mathcal{R}}$ using the randomness (R_1, R_2) , respectively, the probability is taken over random choices of M and r . When assuming G is chosen at random, the probability can be expressed as

$$\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)] = \Pr_{r \leftarrow \psi_{\mathcal{R}}, M \leftarrow \psi_{\mathcal{M}}, G}[c_1 = \text{Enc}(pk, M; r) \wedge c_2 = G(M) \oplus m].$$

Using the Law of Total Probability, we consider all possible values of $r_i \in \mathcal{R}$ and $M_j \in \mathcal{M}$ into conditions:

$$\begin{aligned} & \Pr_{r \leftarrow \psi_{\mathcal{R}}, M \leftarrow \psi_{\mathcal{M}}, G}[c_1 = \text{Enc}(pk, M; r) \wedge c_2 = G(M) \oplus m] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_{M \leftarrow \psi_{\mathcal{M}}, G}[c_1 = \text{Enc}(pk, M; r) \wedge c_2 = G(M) \oplus m \mid r = r_i] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_{M \leftarrow \psi_{\mathcal{M}}, G}[c_1 = \text{Enc}(pk, M; r_i) \wedge c_2 = G(M) \oplus m] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \Pr_G[c_1 = \text{Enc}(pk, M_j; r_i) \wedge c_2 = G(M_j) \oplus m] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i]. \end{aligned}$$

By the random oracle assumption, $G(M_j)$ is uniform and the distribution of $G(M_j) \oplus m$ is identical to that of a uniformly random $\tilde{c} \leftarrow \{0, 1\}^\lambda$.

$$\begin{aligned}
& \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \Pr_G[c_1 = \text{Enc}(pk, M_j; r_i) \wedge c_2 = G(M_j) \oplus m] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \Pr_{u \leftarrow \{0,1\}^\lambda}[c_1 = \text{Enc}(pk, M_j; r_i) \wedge c_2 = u \oplus m] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[c_1 = \text{Enc}(pk, M_j; r_i) \wedge c_2 = \tilde{c}] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i].
\end{aligned}$$

Now, we leverage independence of $c_1 = \text{Enc}(pk, M_j; r_i)$ and $c_2 = \tilde{c}$. To facilitate our analysis, we define two indicator functions $\mathbf{I}(pk, M, r, c_1) = \llbracket c_1 == \text{Enc}(pk, M; r) \rrbracket$ and $\mathbf{I}'(\tilde{c}, c_2) = \llbracket c_2 == \tilde{c} \rrbracket$. Then,

$$\begin{aligned}
& \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[c_1 = \text{Enc}(pk, M_j; r_i) \wedge c_2 = \tilde{c}] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_j \in \mathcal{M}} \sum_{\tilde{c}_k \in \{0,1\}^\lambda} \left(\mathbf{I}(pk, M_j, r_i, c_1) \cdot \mathbf{I}'(\tilde{c}_k, c_2) \cdot \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k] \right) \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\
&= \left(\sum_{\tilde{c}_k \in \{0,1\}^\lambda} \mathbf{I}'(\tilde{c}_k, c_2) \cdot \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k] \right) \cdot \sum_{M_j \in \mathcal{M}} \sum_{r_i \in \mathcal{R}} \mathbf{I}(pk, M_j, r_i, c_1) \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \\
&= \left(\sum_{\tilde{c}_k \in \{0,1\}^\lambda} \mathbf{I}'(\tilde{c}_k, c_2) \cdot \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k] \right) \cdot \sum_{M_j \in \mathcal{M}} \Pr_{r \leftarrow \psi_{\mathcal{R}}}[c_1 = \text{Enc}(pk, M_j; r)] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j].
\end{aligned}$$

Since $\sum_{\tilde{c}_k \in \{0,1\}^\lambda} \mathbf{I}'(\tilde{c}_k, c_2) = 1$ holds for exactly one $\tilde{c}_k \in \{0, 1\}^\lambda$, $\Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k]$ is counted only once. Considering $\Pr_{r \leftarrow \psi_{\mathcal{R}}}[c_1 = \text{Enc}(pk, M_j; r)]$ as the γ -spreadness of PKE for a message M_j , we can use the fact $\Pr_{r \leftarrow \psi_{\mathcal{R}}}[c_1 = \text{Enc}(pk, M_j; r)] \leq 2^{-\gamma}$ for a specific c_1 and each M_j . Therefore, the value $2^{-\gamma}$ can be treated as a constant, and the sum of the probabilities, $\sum_{M_j \in \mathcal{M}} \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] = 1$.

Consequently, the γ' -spreadness of PKE' is derived as follows:

$$\begin{aligned}
\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)] &= \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k] \cdot \sum_{M_j \in \mathcal{M}} \Pr_{r \leftarrow \psi_{\mathcal{R}}}[c_1 = \text{Enc}(pk, M_j; r)] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \\
&\leq \Pr_{\tilde{c} \leftarrow \{0,1\}^\lambda}[\tilde{c} = \tilde{c}_k] \cdot \sum_{M_j \in \mathcal{M}} 2^{-\gamma} \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \\
&= 2^{-\lambda} \cdot 2^{-\gamma} \cdot \sum_{M_j \in \mathcal{M}} \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \\
&= 2^{-\lambda-\gamma}.
\end{aligned}$$

Since the above analysis holds for any (pk, sk) and any $m \in \mathcal{M}$, the γ' -spreadness of PKE' is obtained as $\gamma' = \gamma + \lambda$ as required. $\square$

Corollary 4.2. If PKE is weakly γ -spread and G is modeled as a random oracle, then $\text{PKE}' = \text{ACWC}_0[\text{PKE}, G]$ is weakly γ' -spread with

$$\gamma' = \gamma + \lambda,$$

where λ is the message length of PKE'.

Proof. The weak γ' -spreadness of PKE' is defined as the expectation of the maximum probability of $c = \text{Enc}'(pk, m; R)$ over the key pairs $(pk, sk) \leftarrow \text{Gen}(1^n)$. For each $pk_i \in \mathcal{PK}$, where $\mathcal{PK}$ denotes the set of all possible public keys, the maximum probability of $c_1 = \text{Enc}(pk_i, M; r) \wedge c_2 = G(M) \oplus m$ is determined by the distribution of r , M , and $G(M)$ over the random choice of G . This implies that the maximum probability of $c = \text{Enc}'(pk, m; R)$ is identical for each pk_i . Therefore, the maximum probability can be treated as a constant value when computing the expected probability over the distribution of pk :

$$\begin{aligned}
& \mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{R \leftarrow \mathcal{R}', G} [c = \text{Enc}'(pk, m; R)] \right] \\
&= \mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{r \leftarrow \psi_{\mathcal{R}}, M \leftarrow \psi_{\mathcal{M}}, G} [c_1 = \text{Enc}'(pk, M; r) \wedge c_2 = G(M) \oplus m] \right] \\
&= \sum_{pk_i \in \mathcal{PK}} \max_{\substack{m \in \mathcal{M} \\ c \in \mathcal{C}}} \Pr_{r \leftarrow \psi_{\mathcal{R}}, M \leftarrow \psi_{\mathcal{M}}, G} [c_1 = \text{Enc}'(pk_i, M; r) \wedge c_2 = G(M) \oplus m] \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&\leq \sum_{pk_i \in \mathcal{PK}} 2^{-\gamma'} \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&= 2^{-\gamma'} \cdot \sum_{pk_i \in \mathcal{PK}} \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&= 2^{-\gamma'}.
\end{aligned}$$

□

4.2 New Proof on the γ' -Spreadness of $\text{PKE}' = \text{ACWC}_1[\text{PKE}, \text{GOTP}, G]$

A ciphertext c generated by ACWC_1 is $c = \text{Enc}'(pk, m; (R_1, R_2)) = \text{Enc}(pk, M_1 \parallel \text{Encode}(m, G(M_1)); r)$, where M_1 and r are sampled from $\psi_{\mathcal{M}_1}$ and $\psi_{\mathcal{R}}$ using randomness R_1 and R_2 , respectively. As in ACWC_0 , for a fixed message $m \in \mathcal{M}'$, the message $M_1 \parallel \text{Encode}(m, G(M_1))$ of PKE is changed depending on the choice of M_1 . Under the assumption that G is a random oracle and GOTP satisfies message-hiding, the encoded component $M_2 := \text{Encode}(m, G(M_1))$ is distributed according to $\psi_{\mathcal{M}_2}$, since $G(M_1)$ can be viewed as sampled from $\psi_{\mathcal{U}}$ and the message-hiding property renders M_2 independent of m . We then apply the law of total probability over all choices of $M_1 \parallel M_2$ and r . In this context, the γ' -spreadness of PKE' is analyzed as the maximum probability $\Pr[c = \text{Enc}(pk, M_1 \parallel M_2; r)]$, based on the sampling probabilities of $M_1 \leftarrow \psi_{\mathcal{M}_1}$, $M_2 \leftarrow \psi_{\mathcal{M}_2}$, and $r \leftarrow \psi_{\mathcal{R}}$ in PKE . To organize the contribution of all tuples (M_1, M_2, r) that produce c , we introduce an indicator function for matching, and then, when summing over r , the resulting quantity is upper-bounded by the γ -spreadness of the base PKE in the same manner as in ACWC_0 .

Theorem 4.3. If PKE is γ -spread, GOTP satisfies message-hiding, and G is modeled as a random oracle, then the transformed scheme $\text{PKE}' = \text{ACWC}_1[\text{PKE}, \text{GOTP}, G]$ is γ -spread.

Proof. For a fixed key (pk, sk) and a fixed message m in PKE' , we consider the probability $\Pr_{R \leftarrow \mathcal{R}', G} [c = \text{Enc}'(pk, m; R)]$ for every possible ciphertext c where $R = (R_1, R_2)$. By the fact that M_1 and r are sampled according to $\psi_{\mathcal{M}_1}$ and $\psi_{\mathcal{R}}$ with the randomness R_1 and R_2 respectively, the probability is taken over random choice of M_1 and r . When assuming G is a random oracle, the probability can be expressed as

$$\Pr_{R \leftarrow \mathcal{R}', G} [c = \text{Enc}'(pk, m; R)] = \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, r \leftarrow \psi_{\mathcal{R}}, G} [c = \text{Enc}(pk, M_1 \parallel \text{Encode}(m, G(M_1)); r)].$$

We leverage the Law of Total Probability to consider all possible values of $r_i \in \mathcal{R}$ and $M_{1,i} \in \mathcal{M}_1$ as conditions:

$$\begin{aligned}
& \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, r \leftarrow \psi_{\mathcal{R}}, G} [c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r)] \\
&= \sum_{r_i \in \mathcal{R}} \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, G} [c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r) \mid r = r_i] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, G} [c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_G [c = \text{Enc}(pk, M_{1,j} || \text{Encode}(m, G(M_{1,j})); r_i)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i].
\end{aligned}$$

By the fact that $G(M_{1,j})$ is $\psi_{\mathcal{U}}$ -distributed and the message hiding property of GOTP holds for $G(M_{1,j})$, the output $M_2 = \text{Encode}(m, G(M_{1,j}))$ is $\psi_{\mathcal{M}_2}$ -distributed over random choice of G .

$$\begin{aligned}
& \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_G [c = \text{Enc}(pk, M_{1,j} || \text{Encode}(m, G(M_{1,j})); r_i)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{u \leftarrow \psi_{\mathcal{U}}} [c = \text{Enc}(pk, M_{1,j} || \text{Encode}(m, u); r_i)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}} [c = \text{Enc}(pk, M_{1,j} || M_2; r_i)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i].
\end{aligned}$$

Next, we consider all possible combinations of (r, M_1, M_2) which is mapped to a specific c . Introducing an indicator $\mathbf{I}(pk, M_1, M_2, r, c) = \llbracket c == \text{Enc}(pk, M_1 || M_2; r) \rrbracket$, the sum can be rewritten as follows :

$$\begin{aligned}
& \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}} [c = \text{Enc}(pk, M_{1,j} || M_2; r_i)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{r_i \in \mathcal{R}} \sum_{M_{1,j} \in \mathcal{M}_1} \left(\sum_{M_{2,k} \in \mathcal{M}_2} \mathbf{I}(pk, M_{1,j}, M_{2,k}, r_i, c) \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}} [M_2 = M_{2,k}] \right) \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \\
&= \sum_{M_{2,k} \in \mathcal{M}_2} \sum_{M_{1,j} \in \mathcal{M}_1} \left(\sum_{r_i \in \mathcal{R}} \mathbf{I}(pk, M_{1,j}, M_{2,k}, r_i, c) \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}} [r = r_i] \right) \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}} [M_2 = M_{2,k}] \\
&= \sum_{M_{2,k} \in \mathcal{M}_2} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, M_{1,j} || M_{2,k}; r)] \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}} [M_1 = M_{1,j}] \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}} [M_2 = M_{2,k}].
\end{aligned}$$

Considering $\Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, M_{1,j} || M_{2,k}; r)]$ over all messages $M_{1,j} || M_{2,k}$, the maximum probability is upper-bounded by $2^{-\gamma}$ by the γ -spreadness of PKE. Since $2^{-\gamma}$ is treated as a constant, the sum of the probabilities of M_1 and M_2 becomes 1, respectively. Then, the γ' -spreadness of PKE' is upper-bounded

as:

$$\begin{aligned}
\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)] &= \sum_{M_{2,k} \in \mathcal{M}_2} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{r \leftarrow \psi_{\mathcal{R}}}[c = \text{Enc}(pk, M_{1,j} || M_{2,k}; r)] \\
&\quad \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}}[M_1 = M_{1,j}] \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}}[M_2 = M_{2,k}] \\
&\leq \sum_{M_{2,k} \in \mathcal{M}_2} \sum_{M_{1,j} \in \mathcal{M}_1} 2^{-\gamma} \cdot \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}}[M_1 = M_{1,j}] \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}}[M_2 = M_{2,k}] \\
&= 2^{-\gamma} \cdot \sum_{M_{2,k} \in \mathcal{M}_2} \sum_{M_{1,j} \in \mathcal{M}_1} \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}}[M_1 = M_{1,j}] \cdot \Pr_{M_2 \leftarrow \psi_{\mathcal{M}_2}}[M_2 = M_{2,k}] \\
&= 2^{-\gamma}.
\end{aligned}$$

Since this holds for any (pk, sk) and any $m \in \mathcal{M}$, the γ' -spreadness of PKE' is obtained as $\gamma' = \gamma$ as required. $\square$

Corollary 4.4. If PKE is weakly γ -spread, GOTP has the message hiding property, and G is modeled as a random oracle, then $\text{PKE}' = \text{ACWC}_1[\text{PKE}, \text{GOTP}, G]$ is also weakly γ -spread.

Proof. For each $pk_i \in \mathcal{PK}$, where $\mathcal{PK}$ denotes the set of all possible public keys, the maximum probability of $c = \text{Enc}(pk_i, M_1 || \text{Encode}(m, G(M_1)); r)$ is determined by the distribution of r , M_1 and $\text{Encode}(m, G(M_1))$ over the random choice of G . This implies that the maximum probability of $c = \text{Enc}'(pk, m; R)$ is identical for each pk_i . Hence, the maximum probability can be treated as a constant when computing the expected value over the distribution of pk :

$$\begin{aligned}
&\mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)] \right] \\
&= \mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, r \leftarrow \psi_{\mathcal{R}}, G}[c = \text{Enc}(pk, M_1 || \text{Encode}(m, G(M_1)); r)] \right] \\
&= \sum_{pk_i \in \mathcal{PK}} \max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{M_1 \leftarrow \psi_{\mathcal{M}_1}, r \leftarrow \psi_{\mathcal{R}}, G}[c = \text{Enc}(pk_i, M_1 || \text{Encode}(m, G(M_1)); r)] \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)}[pk = pk_i] \\
&\leq \sum_{pk_i \in \mathcal{PK}} 2^{-\gamma} \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)}[pk = pk_i] \\
&= 2^{-\gamma} \cdot \sum_{pk_i \in \mathcal{PK}} \Pr_{pk \leftarrow \text{Gen}(1^n)}[pk = pk_i] \\
&= 2^{-\gamma}.
\end{aligned}$$

$\square$

4.3 New Proof on the γ' -Spreadness of $\text{PKE}' = \text{ACWC}_2[\text{PKE}, \text{SOTP}, G]$

Building on the framework of [14], we retain its high-level structure but derive a refined and tighter upper bound on the γ' -spreadness of PKE' in Theorem 4.5.

A ciphertext in ACWC_2 is of the form $c = \text{Enc}'(pk, m; R) = \text{Enc}(pk, \text{Encode}(m, G(r)); r)$, where r is sampled from $\psi_{\mathcal{R}}$ according to R . For a fixed $m \in \mathcal{M}'$, the encoded message $M = \text{Encode}(m, G(r))$ for PKE is changed depending on the choice of r . Under the random oracle model for G and assuming

SOTP satisfies message-hiding, M is distributed as $\psi_{\mathcal{M}}$. To consider all possible cases of M and r , we also leverage the Law of Total Probability. As a result, the γ' -spreadness of PKE' is analyzed as the maximum probability $\Pr[c = \text{Enc}(pk, M; r)]$, based on the sampling probabilities of $M \leftarrow \psi_{\mathcal{M}}$, and $r \leftarrow \psi_{\mathcal{R}}$ in PKE . We define an indicator function to account for the probabilities of all possible combinations (M, r) that match a specific ciphertext c . Over the randomness sampled according to $r \leftarrow \psi_{\mathcal{R}}$, the sum of the probabilities regarding the indicator function can be upper-bounded by the γ -spreadness of PKE .

Theorem 4.5. If PKE is γ -spread, SOTP has the message hiding property, and G is modeled as a random oracle, then $\text{PKE}' = \text{ACWC}_2[\text{PKE}, \text{SOTP}, G]$ is also γ -spread.

Proof. For a fixed key (pk, sk) and a fixed message m in PKE' , we consider the probability $\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)]$ for every possible ciphertext c . The probability is taken over a random choice of r , assuming G is a random oracle. When r is sampled as $r \leftarrow \psi_{\mathcal{R}}$ with the randomness $R \leftarrow \mathcal{R}'$, the probability can be expressed as

$$\Pr_{R \leftarrow \mathcal{R}', G}[c = \text{Enc}'(pk, m; R)] = \Pr_{r \leftarrow \psi_{\mathcal{R}}, G}[c = \text{Enc}(pk, \text{Encode}(m, G(r)); r)].$$

We leverage the Law of Total Probability to consider all possible values of $r_i \in \mathcal{R}$ as conditions:

$$\begin{aligned} & \Pr_{r \leftarrow \psi_{\mathcal{R}}, G}[c = \text{Enc}(pk, \text{Encode}(m, G(r)); r)] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_G[c = \text{Enc}(pk, \text{Encode}(m, G(r)); r) \mid r = r_i] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_G[c = \text{Enc}(pk, \text{Encode}(m, G(r_i)); r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i]. \end{aligned}$$

By the fact that $G(r_i)$ is $\psi_{\mathcal{U}}$ -distributed and the message hiding property of SOTP holds, the output $M = \text{Encode}(m, G(r_i))$ is $\psi_{\mathcal{M}}$ -distributed over random choice of G .

$$\begin{aligned} & \sum_{r_i \in \mathcal{R}} \Pr_G[c = \text{Enc}(pk, \text{Encode}(m, G(r_i)); r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_{u \leftarrow \psi_{\mathcal{U}}}[c = \text{Enc}(pk, \text{Encode}(m, u); r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \Pr_{M \leftarrow \psi_{\mathcal{M}}}[c = \text{Enc}(pk, M; r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i]. \end{aligned}$$

Next, we consider all possible combinations of (r, M) which is mapped to a specific c . To facilitate our analysis, we define an indicator function $\mathbf{I}(pk, M, r, c) = \llbracket c == \text{Enc}(pk, M; r) \rrbracket$. Then,

$$\begin{aligned} & \sum_{r_i \in \mathcal{R}} \Pr_{M \leftarrow \psi_{\mathcal{M}}}[c = \text{Enc}(pk, M; r_i)] \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{r_i \in \mathcal{R}} \left(\sum_{M_j \in \mathcal{M}} \mathbf{I}(pk, M_j, r_i, c) \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \right) \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \\ &= \sum_{M_j \in \mathcal{M}} \left(\sum_{r_i \in \mathcal{R}} \mathbf{I}(pk, M_j, r_i, c) \cdot \Pr_{r \leftarrow \psi_{\mathcal{R}}}[r = r_i] \right) \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j] \\ &= \sum_{M_j \in \mathcal{M}} \Pr_{r \leftarrow \psi_{\mathcal{R}}}[c = \text{Enc}(pk, M_j; r)] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}}[M = M_j]. \end{aligned}$$

By the γ -spreadness of PKE, for each fixed ciphertext c and message M_j , the maximum probability $\Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, M_j; r)]$ is upper-bounded by $2^{-\gamma}$. Treating $2^{-\gamma}$ as a constant and summing over $M_j \leftarrow \psi_{\mathcal{M}}$ yields $\sum_{M_j} \Pr_M [M = M_j] = 1$. Hence the γ' -spreadness of PKE' is bounded by :

$$\begin{aligned} \Pr_{R \leftarrow \mathcal{R}', G} [c = \text{Enc}'(pk, m; R)] &= \sum_{M_j \in \mathcal{M}} \Pr_{r \leftarrow \psi_{\mathcal{R}}} [c = \text{Enc}(pk, M_j; r)] \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}} [M = M_j] \\ &\leq \sum_{M_j \in \mathcal{M}} 2^{-\gamma} \cdot \Pr_{M \leftarrow \psi_{\mathcal{M}}} [M = M_j] \\ &= 2^{-\gamma} \cdot \sum_{M_j \in \mathcal{M}} \Pr_{M \leftarrow \psi_{\mathcal{M}}} [M = M_j] \\ &= 2^{-\gamma}. \end{aligned}$$

Since the above analysis holds for any (pk, sk) and any $m \in \mathcal{M}$, the γ' -spreadness of PKE' is obtained as $\gamma' = \gamma$, as required. $\square$

Corollary 4.6. If PKE is weakly γ -spread, SOTP has the message hiding property, and G is modeled as a random oracle, then $\text{PKE}' = \text{ACWC}_2[\text{PKE}, \text{SOTP}, G]$ is also weakly γ -spread.

Proof. By the definition of the weak γ -spreadness, we consider the expectation of the maximum probability for each $pk_i \in \mathcal{PK}$. The maximum probability of $c = \text{Enc}(pk_i, \text{Encode}(m, G(r)); r)$ is determined by the distribution of r and $\text{Encode}(m, G(r))$ over the random choice of G. This implies that the maximum probability is identical for each pk_i . Therefore, the maximum probability can be treated as a constant value when computing the expected probability for the distribution of pk :

$$\begin{aligned} &\mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{R \leftarrow \mathcal{R}', G} [c = \text{Enc}'(pk, m; R)] \right] \\ &= \mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{r \leftarrow \psi_{\mathcal{R}}, G} [c = \text{Enc}(pk, \text{Encode}(m, G(r)); r)] \right] \\ &= \sum_{pk_i \in \mathcal{PK}} \max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr_{r \leftarrow \psi_{\mathcal{R}}, G} [c = \text{Enc}(pk_i, \text{Encode}(m, G(r)); r)] \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\ &\leq \sum_{pk_i \in \mathcal{PK}} 2^{-\gamma} \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\ &= 2^{-\gamma} \cdot \sum_{pk_i \in \mathcal{PK}} \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\ &= 2^{-\gamma}. \end{aligned}$$

$\square$

5 Applications to NTRU-Based PKE Schemes

5.1 Notations

Let $R_q = \mathbb{Z}_q[x] / \langle f(x) \rangle$ be a ring, where q is an odd prime modulus and $f(x)$ is a cyclotomic polynomial of degree n . We define $y = x \bmod q$ as the unique integer $y \in \{-(q-1)/2, \dots, (q-1)/2\}$ that satisfies $q|y-x$. For a polynomial $\mathbf{a} \in R_q$, we define ' $\mathbf{a} \bmod q$ ' to compute the modular operation for all coefficients

of $\mathbf{a}$. For a distribution D , we define ' $\mathbf{a} \leftarrow D^n$ ' to sample a polynomial $\mathbf{a} \in R_q$ whose coefficients are sampled according to D .

5.2 GenNTRU $[D^n]$ as PKE

We assume that R_q is defined for an appropriate cyclotomic polynomial $f(x)$ with degree n . Then, Figure 4 shows GenNTRU $[D^n] = (\text{Gen}, \text{Enc}, \text{Dec}, \text{RRec})$ as an underlying PKE, where D is a distribution. GenNTRU $[D^n]$ is randomness-recoverable under the condition that $\mathbf{h}$ is invertible, and RRec is a randomness-recovery algorithm. We notice that RRec is necessary for only ACWC $_2$.

<u>Gen(1^n)</u>	<u>Enc($\mathbf{h}, M \leftarrow D^n; \mathbf{r} \leftarrow D^n$)</u>
1: $\mathbf{f}', \mathbf{g} \leftarrow D^n$	1: return $\mathbf{c} := \mathbf{h}\mathbf{r} + M \bmod q$
2: $\mathbf{f} := 3\mathbf{f}' + 1$	
3: if $\mathbf{f}, \mathbf{g}$ are not invertible in R_q , restart	<u>Dec($\mathbf{f}, \mathbf{c}$)</u>
4: $\mathbf{h} := 3\mathbf{g}\mathbf{f}^{-1} \bmod q$	1: return $\mathbf{m} := (\mathbf{c}\mathbf{f} \bmod q) \bmod 3$
5: return $(pk, sk) = (\mathbf{h}, \mathbf{f})$	<u>RRec($\mathbf{h}, \mathbf{m}, \mathbf{c}$)</u>
	1: return $\mathbf{r} := (\mathbf{c} - \mathbf{m})\mathbf{h}^{-1} \bmod q$

Figure 4: GenNTRU $[D^n]$

A desirable feature of GenNTRU $[D^n]$ is that its γ -spreadness admits a straightforward statistical characterization: for any fixed key pair (pk, sk) and message M , there is exactly one randomness $\mathbf{r}$ satisfying $\mathbf{c} = \text{Enc}(\mathbf{h}, M; \mathbf{r})$ for a given ciphertext $\mathbf{c} \in \mathcal{C}$ [13]. Consequently, to maximize the probability that M meets $\mathbf{c}$, we can simply consider the maximum probability $\hat{p}$ of sampling a randomness $\mathbf{r}$ over a distribution D . Since this holds for any key pair (pk, sk) and any message M (sampled from its message space $\mathcal{M}$), the γ -spreadness of GenNTRU $[D^n]$ becomes $\gamma = -\log \hat{p}$.

5.3 NTRU-C := ACWC $_0$ [GenNTRU $[\bar{\psi}_2^n]$, G]

Let $\bar{\psi}_2$ be the distribution that is obtained by computing $(b_0 + b_1 - b_2 - b_3) \bmod 3$ for uniformly random bits $\{b_i\}_{i=0}^3$. Notice that $\bar{\psi}_2$ samples $-1, 0, 1$ with probabilities $5/16, 6/16, 5/16$, respectively. For a hash function $G : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$, NTRU-C [5] is obtained by applying ACWC $_0$ to GenNTRU $[\bar{\psi}_2^n]$, which is shown in Figure 5.

<u>Gen'(1ⁿ)</u>	<u>Dec'(f, c)</u>
1: $\mathbf{f}', \mathbf{g} \leftarrow \overline{\psi}_2^n$	1: $M = (\mathbf{c}_1 \mathbf{f} \bmod q) \bmod 3$
2: $\mathbf{f} = 3\mathbf{f}' + 1$	2: $\mathbf{m} = \mathbf{c}_2 \oplus \mathbf{G}(M)$
3: if $\mathbf{f}, \mathbf{g}$ are not invertible in R_q , restart	3: return $\mathbf{m}$
4: $\mathbf{h} = 3\mathbf{g}\mathbf{f}^{-1} \bmod q$	
5: return $(pk, sk) = (\mathbf{h}, \mathbf{f})$	
<u>Enc'($\mathbf{h}, \mathbf{m} \in \{0, 1\}^\lambda; (R_1, R_2) \in \mathcal{R}'$)</u>	
1: $M \leftarrow \overline{\psi}_2^n$ using the randomness R_1	
2: $\mathbf{r} \leftarrow \overline{\psi}_2^n$ using the randomness R_2	
3: $\mathbf{c}_1 = \mathbf{h}\mathbf{r} + M \bmod q$	
4: $\mathbf{c}_2 = \mathbf{m} \oplus \mathbf{G}(M)$	
5: return $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$	

Figure 5: NTRU-C

Lemma 5.1. (Spreadness) $\text{GenNTRU}[\overline{\psi}_2^n]$ is $n \log(\frac{8}{3})$ -spread.

Proof. Since the maximum probability of sampling $\mathbf{r}$ over $\overline{\psi}_2^n$ is $(\frac{3}{8})^n$, we see that $\text{GenNTRU}[\overline{\psi}_2^n]$ is $n \log(\frac{8}{3})$ -spread. $\square$

Lemma 5.2. (Weak spreadness) $\text{GenNTRU}[\overline{\psi}_2^n]$ is weakly $n \log(\frac{8}{3})$ -spread.

Proof. The γ -spreadness of $\text{GenNTRU}[\overline{\psi}_2^n]$ only depends on the maximum probability of sampling $\mathbf{r}$ from $\overline{\psi}_2^n$, and thus the maximum probability remains the same across all generated (pk, sk) . Based on this, the expected maximum probability over (pk, sk) is given as follows:

$$\begin{aligned}
\mathbb{E} \left[\max_{\mathbf{m} \in \mathcal{M}, \mathbf{c} \in \mathcal{C}} \Pr_{\mathbf{r} \leftarrow \overline{\psi}_2^n} [\mathbf{c} = \text{Enc}(pk, \mathbf{m}; \mathbf{r})] \right] &= \sum_{pk_i \in \mathcal{PK}} \max_{\substack{\mathbf{m} \in \mathcal{M} \\ \mathbf{c} \in \mathcal{C}}} \Pr_{\mathbf{r} \leftarrow \overline{\psi}_2^n} [\mathbf{c} = \text{Enc}(pk_i, \mathbf{m}; \mathbf{r})] \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&\leq \sum_{pk_i \in \mathcal{PK}} 2^{-\gamma} \cdot \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&= \left(\frac{3}{8}\right)^n \cdot \sum_{pk_i \in \mathcal{PK}} \Pr_{pk \leftarrow \text{Gen}(1^n)} [pk = pk_i] \\
&= \left(\frac{3}{8}\right)^n.
\end{aligned}$$

This concludes that $\text{GenNTRU}[\overline{\psi}_2^n]$ is weakly $n \log(\frac{8}{3})$ -spread. $\square$

Lemma 5.3. NTRU-C is (weakly) $(n \log(\frac{8}{3}) + \lambda)$ -spread.

Proof. Based on the above result that $\text{GenNTRU}[\overline{\psi}_2^n]$ is (weakly) $n \log(\frac{8}{3})$ -spread, Theorem 4.1 and Corollary 4.2 give the (weak) γ' -spreadness of $\text{NTRU-C} = \text{ACWC}_0[\text{GenNTRU}[\overline{\psi}_2^n], \mathbf{G}]$ as $\gamma' = n \log(\frac{8}{3}) + \lambda$. $\square$

5.4 NTRU-B := ACWC₁[GenNTRU[U_3^n], GOTP, G]

Denote by U_3 the uniform distribution on $\{-1, 0, 1\}$, and instantiate the hash function $G : \{0, 1\}^* \rightarrow \{-1, 0, 1\}^\lambda$ so that its outputs follow U_3^λ . The concrete instantiation of GOTP is given in Figure 6, where message-hiding is immediate from sampling $u \leftarrow U_3^\lambda$. Moreover, GOTP requires the randomness-hiding, i.e., each $x \in \mathcal{M}' = \{-1, 0, 1\}^\lambda$ must be sampled according to U_3^λ . Applying ACWC₁ to GenNTRU[U_3^n] with these primitives yields NTRU-B [5], as illustrated in Figure 7.

<u>Encode($x \in \mathcal{M}' = \{-1, 0, 1\}^\lambda, u \leftarrow U_3^\lambda$)</u>	<u>Inv($y \in \mathcal{M}_2 = \{-1, 0, 1\}^\lambda, u \leftarrow U_3^\lambda$)</u>
1: $y = x + u \pmod 3$	1: $x = y - u \pmod 3$
2: return y	2: return x

Figure 6: GOTP instantiation

<u>Gen'(1ⁿ)</u>	<u>Dec'(f, c)</u>
1: $\mathbf{f}', \mathbf{g} \leftarrow U_3^n$	1: $M_1 M_2 = (\mathbf{c}\mathbf{f} \pmod q) \pmod 3$
2: $\mathbf{f} = 3\mathbf{f}' + 1$	2: $\mathbf{m} = \text{Inv}(M_2, G(M_1)):$
3: if $\mathbf{f}, \mathbf{g}$ are not invertible in R_q , restart	– $\mathbf{u} = G(M_1)$
4: $\mathbf{h} = 3\mathbf{g}\mathbf{f}^{-1} \pmod q$	– $\mathbf{m} = M_2 - \mathbf{u} \pmod 3$
5: return $(pk, sk) = (\mathbf{h}, \mathbf{f})$	3: return $\mathbf{m}$
<u>Enc'($\mathbf{h}, \mathbf{m} \leftarrow U_3^\lambda; (R_1, R_2) \in \mathcal{R}'$)</u>	
1: $\mathbf{r} \leftarrow U_3^n$ using the randomness R_1	
2: $M_1 \leftarrow U_3^{n-\lambda}$ using randomness R_2	
3: $M_2 = \text{Encode}(\mathbf{m}, G(M_1)):$	
– $\mathbf{u} = G(M_1)$	
– $M_2 = \mathbf{m} + \mathbf{u} \pmod 3$	
4: $M = M_1 M_2$	
5: $\mathbf{c} = \mathbf{h}\mathbf{r} + M \pmod q$	
6: return $\mathbf{c}$	

Figure 7: NTRU-B

Lemma 5.4. (Spreadness) GenNTRU[U_3^n] is $n \log 3$ -spread.

Proof. Since the distribution of $\mathbf{r}$ is uniform over $\{-1, 0, 1\}$, the maximum probability of sampling $\mathbf{r}$ over U_3^n is $(\frac{1}{3})^n$. Therefore, we see that GenNTRU[U_3^n] is $n \log 3$ -spread. $\square$

Lemma 5.5. (Weak spreadness) GenNTRU[U_3^n] is weakly $n \log 3$ -spread.

Proof. The γ -spreadness of GenNTRU[U_3^n] is determined solely by the maximum probability of sampling $\mathbf{r}$ from U_3^n , which remains invariant across all key pairs (pk, sk) . An argument similar to Lemma 5.2 can be made. $\square$

Lemma 5.6. NTRU-B is (weakly) $n \log 3$ -spread.

Proof. Based on the above result that GenNTRU[U_3^n] is (weakly) $n \log 3$ -spread, Theorem 4.3 and Corollary 4.4 give the (weak) γ' -spreadness of NTRU-B = ACWC₁[GenNTRU[U_3^n], GOTP, G] as $\gamma' = n \log 3$. $\square$

5.5 NTRU+ := ACWC₂[GenNTRU[ψ_1^n], SOTP, G]

Let ψ_1 denote the centered binomial distribution that is obtained by computing $(b_0 - b_1)$ for two uniformly random bits b_0 and b_1 . Notice that ψ_1 samples $-1, 0, 1$ with the probabilities $1/4, 1/2, 1/4$, respectively. Let G be a hash function $G : \{0, 1\}^* \rightarrow \{0, 1\}^{2n}$ whose output is uniform over $\{0, 1\}^{2n}$. The instantiation of SOTP is specified in Figure 8. The message-hiding property follows from the one-time pad $x \oplus u_1$ with $u_1 \leftarrow \{0, 1\}^n$. Also, for another uniformly-random n -bit u_2 , the rigidity of SOTP comes from the fact that $x = (y + u_2) \oplus u_1$ implies $y = (x \oplus u_1) - u_2$, as long as $y + u_2 \in \{0, 1\}^n$. Figure 9 shows NTRU+ [13] that is obtained by applying ACWC₂ to GenNTRU[ψ_1^n].

Encode($x \in \mathcal{M}' = \{0, 1\}^n, u \leftarrow U^{2n}$)	Inv($y \in \mathcal{M} = \{-1, 0, 1\}^n, u \in \{0, 1\}^{2n}$)
1: $u = (u_1, u_2) \in \{0, 1\}^n \times \{0, 1\}^n$	1: $u = (u_1, u_2) \in \{0, 1\}^n \times \{0, 1\}^n$
2: $y = (x \oplus u_1) - u_2 \in \{-1, 0, 1\}^n$	2: if $y + u_2 \notin \{0, 1\}^n$, return $\perp$
3: return y	3: Else $x = (y + u_2) \oplus u_1 \in \{0, 1\}^n$
	4: return x

Figure 8: SOTP instantiation

Gen'(1 ⁿ)	Dec'(sk, c)
1: $\mathbf{f}', \mathbf{g} \leftarrow \psi_1^n$	1: $M = (\mathbf{c}\mathbf{f} \bmod q) \bmod 3$
2: $\mathbf{f} = 3\mathbf{f}' + 1$	2: $\mathbf{r} = (\mathbf{c} - M)\mathbf{h}^{-1} \bmod q$
3: if $\mathbf{f}, \mathbf{g}$ are not invertible in R_q , restart	3: $\mathbf{m} = \text{Inv}(M, G(\mathbf{r})) :$
4: $\mathbf{h} = 3\mathbf{g}\mathbf{f}^{-1} \bmod q$	– $G(\mathbf{r}) = (u_1, u_2)$
5: return $(pk, sk) = (\mathbf{h}, \mathbf{f})$	– if $M + u_2 \notin \{0, 1\}^n$, return $\perp$
	– $\mathbf{m} = (M + u_2) \oplus u_1$
	4: return $\mathbf{m}$
Enc'(pk, $\mathbf{m} \in \{0, 1\}^n; R \in \mathcal{R}'$)	
1: $\mathbf{r} \leftarrow \psi_1^n$ using randomness R	
2: $M = \text{Encode}(\mathbf{m}, G(\mathbf{r})) :$	
– $G(\mathbf{r}) = (u_1, u_2)$	
– $M = (\mathbf{m} \oplus u_1) - u_2$	
3: $\mathbf{c} = \mathbf{h}\mathbf{r} + M \bmod q$	
4: return $\mathbf{c}$	

Figure 9: NTRU+

Lemma 5.7. (Spreadness) GenNTRU[ψ_1^n] is n -spread.

Proof. Since sampling $\mathbf{r}$ from ψ_1^n yields a maximum probability of $(\frac{1}{2})^n$, it follows that GenNTRU[ψ_1^n] achieves n -spreadness. $\square$

Lemma 5.8. (Weak spreadness) GenNTRU[ψ_1^n] is weakly n -spread.

Proof. The γ -spreadness of GenNTRU[ψ_1^n] is determined by the maximum probability of sampling $\mathbf{r}$ from ψ_1^n , which remains invariant across all key pairs (pk, sk) . Therefore, an argument similar to Lemma 5.2 can be made. $\square$

Lemma 5.9. NTRU+ is (weakly) n -spread.

Proof. Based on the above result that $\text{GenNTRU}[\psi_1^n]$ is (weakly) n -spread, Theorem 4.5 and Corollary 4.6 give the (weak) γ' -spreadness of $\text{NTRU+} = \text{ACWC}_2[\text{GenNTRU}[\psi_1^n], \text{SOTP}, G]$ as $\gamma' = n$. $\square$

5.6 (Weak) Spreadness on Concrete Parameters

All the (weak) γ' -spreadness of $\{\text{NTRU-C}, \text{NTRU-B}, \text{NTRU+}\}$ are determined by the choice of n and λ in NTRU-C. We choose $n = 576, 768, 864, 1152$ as the degree of cyclotomic trinomials such as $x^n - x^{n/2} + 1$, and we set $\lambda = 256$ as the bit-length of a message only for NTRU-C. Table 1 shows the (weak) γ (or γ')-spreadness of each NTRU-based scheme, as obtained from our refined analysis.

Table 1: (Weak) spreadness calculated from our new spreadness proofs

Scheme	(Weak) ★-spreadness	$\lambda = 256$			
		$n = 576$	$n = 768$	$n = 864$	$n = 1152$
$\text{GenNTRU}[\overline{\psi}_2^n]$	$n \log \left(\frac{8}{3} \right)$	818	1091	1225	1636
$\text{NTRU-C}[5]$	$n \log \left(\frac{8}{3} \right) + \lambda$	1074	1347	1513	1892
$\text{GenNTRU}[U_3^n]$	$n \log 3$	910	1213	1362	1820
$\text{NTRU-B}[5]$					
$\text{GenNTRU}[\psi_1^n]$	n	576	768	864	1152
$\text{NTRU+}[13]$					

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